

PROJECT ASSIGNMENT FOR THE LOGIC COURSE

FUZZY

Informatics Study Program

Final Output: Scientific Article (SINTA Journal Template 3/4/5)

Weight: 40% of Course Grade

1. Task Description

This project aims to enable students to apply Fuzzy Logic to solve real-world problems using real data.

The primary output is a scientific article written using the official template of a national journal accredited with SINTA 3, SINTA 4, or SINTA 5.

2. Learning Outcomes

Students are able to:

- Understand the concept of Fuzzy Logic.
- Choose the appropriate fuzzy method.
- Process real data.
- Implement the method using Python/MATLAB/R.
- Analyze the results.
- Compiling scientific articles.

3. Project Output

Each group is required to submit:

- Articles (.docx and .pdf)
- Source code

- Dataset
- Experimental results
- Presentation slides
- Presentation video (10–15 minutes)

4. Group Requirements

Groups consist of 2–3 students. Each group must choose a different topic.

5. Article Requirements (MANDATORY)

Articles MUST be written using the official template of one of the national journals SINTA 3, SINTA 4, or SINTA 5 that is relevant to the research topic.

Students are required to write directly on the journal template.

6. Reference Journal Provisions (MANDATORY)

Each group must: 1.

Select at least one SINTA 3/4/5 journal.

2. Download the official journal template.

3. Follow all author guidelines.

4. Include the journal name, SINTA ranking, journal URL, and template URL in the report.

7. Research Provisions (MANDATORY)

Project is a research extension.

Students are

required to: • Replicate at least one SINTA journal

article. • Develop research through new datasets, new case studies, comparative methods, parameter optimization, additional variables, or additional evaluations.

Copying research without development is not permitted.

8. Dataset Terms

The dataset must be real data from official sources, for example: • BPS

- BMKG
- One Data Indonesia
- WHO
- World Bank
- Kaggle
- UCI Machine Learning Repository •

Open Government Data

Use of simulation data is only permitted with the approval of the lecturer.

9. Choice of Project Theme A.

Fuzzy Inference System - Mamdani

- Sugeno
- Tsukamoto

B. Fuzzy Forecasting -
Chen FTS

- Song & Chissom
- High Order FTS
- Weighted FTS
- Type-2 FTS

C. Fuzzy Optimization -
Fuzzy Linear Programming - Fuzzy
Goal Programming - Multi-objective
Fuzzy Programming

D. Fuzzy MADM -
Fuzzy AHP -
Fuzzy TOPSIS -
Fuzzy SAW -
Fuzzy VIKOR -
Fuzzy ELECTRE -
Fuzzy PROMETHEE

E. Fuzzy Clustering -
Fuzzy C-Means -
Possibilistic C-Means
- Gustafson-Kessel
- FCM Kernel

10. Project Stages

1. Identify the problem
2. Literature review (≥20 references; ≥70% last five years)
3. Dataset collection
4. Pre-processing
5. Implementation of fuzzy methods
6. Experiments (minimum 3 scenarios)
7. Evaluation
8. Discussion
9. Conclusion

11. Software

Python, MATLAB, or R. Libraries such as scikit-fuzzy, NumPy, Pandas, Matplotlib, PyFTS, SciPy, and Scikit-learn are allowed.

12. Presentation

Duration 15 minutes (10 minutes presentation, in video format).

13. Assessment Rubric

Component	Weight
Identification of problems	10%
Literature review	10%
Dataset quality	10%
Implementation of fuzzy method	25%
Experiments and evaluation	15%
Analysis and discussion	15%
Scientific articles	10%
Presentation	5%

14. Additional Terms

- Maximum script similarity is 20%.
- All code must be executable.
- The use of AI is permitted as an aid, but students must understand and be able to explain the entire contents of the research.
- Articles must be in a ready to submit condition according to the selected journal template.

15. Bonus Value

- Submit to journal: +5 points
- Accepted at national seminar: +7 points
- Accepted/published in the SINTA journal: +10 points (following the maximum course grade limit).